Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two particles start simultaneously from the same point and move along two straight lines, one with uniform velocity v and other with a uniform acceleration a. If α is the angle between the lines of motion of two particles then the least value of relative velocity will be at time given by

(A) $\frac{v}{\alpha}\sin\alpha$

(B) $\frac{v}{a}\cos\alpha$

(C) $\frac{v}{a}\tan\alpha$

(D) $\frac{v}{a}\cot\alpha$

Q.2

In a car race, car A takes t_0 time less to finish than car B and passes the finishing point with a velocity v_0 more than car B. The cars start from rest and travel with constant accelerations a_1 and a_2. Then the ratio $\frac{v_0}{t_0}$ is equal to

(A) $\frac{a_1^2}{a_2}$

(B) $\frac{a_1+a_2}{2}$

(C) $\sqrt{a_1 a_2}$

(D) $\frac{a_2^2}{a_1}$

Q.3

A man holds an umbrella at 30° with the vertical to keep himself dry. He, then, runs at a speed of $10\ \text{ms}^{-1}$ and finds the rain drops to be hitting vertically. Speed of the rain drops w.r.t the running man and w.r.t earth are

(A) $20\ \text{ms}^{-1}, 10\ \text{ms}^{-1}$

(B) $10\ \text{ms}^{-1}, 20\sqrt{3}\ \text{ms}^{-1}$

(C) $10\sqrt{3}\ \text{ms}^{-1}, 20\ \text{ms}^{-1}$

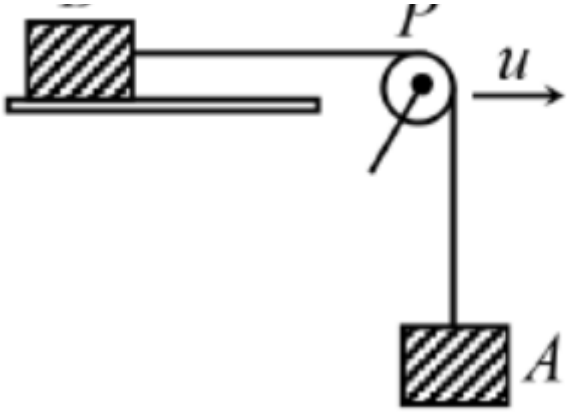
(D) $20\ \text{ms}^{-1}, 10\sqrt{3}\ \text{ms}^{-1}$

Q.4

In the figure, the pulley P moves to the right with a constant speed u. The downward speed of A is v_A and the speed of B to the right is v_{B-}. Which of the following is correct -



@IITJEE_Advanced



(A) $v_B = v_A$

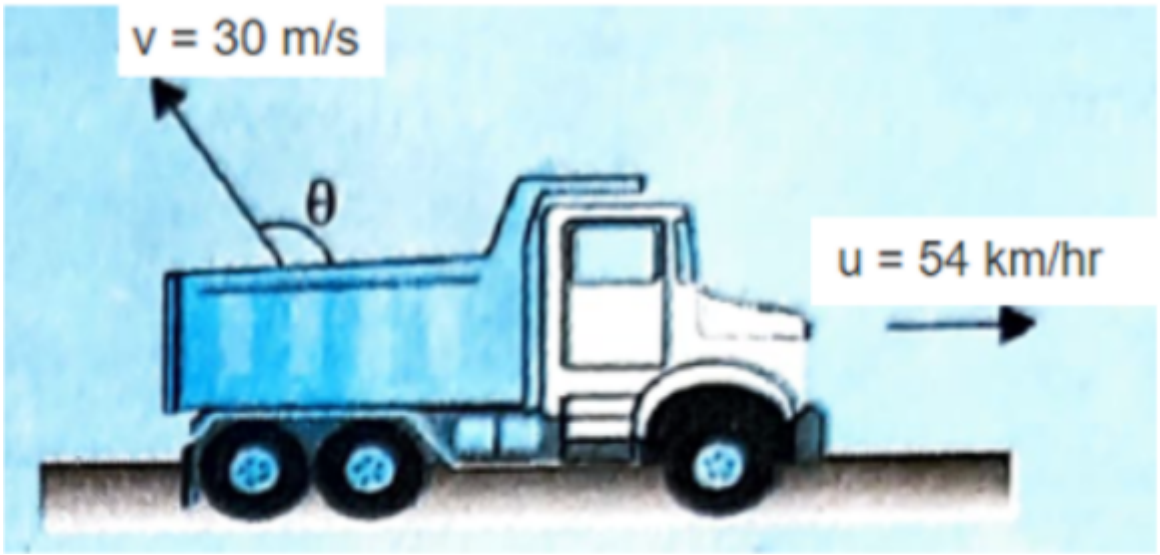
(B) $v_B = u + v_A$

(C) $v_B + u = v_A$

(D) the two blocks have accelerations of the different magnitude.

Q.5

A truck is moving a constant velocity of $u = 54\ \mathrm{km/hr}$. In what direction should a stone be projected up with a velocity of $v = 30\ \mathrm{m/s}$, from the floor of the truck, so as to appear at right angles to the truck, for a person standing on earth?



(A) 60°

(B) 30°

(C) 150°

(D) 120°

Q.6

A bird is flying due east with a velocity of $4\ \mathrm{m\ s^{-1}}$. The wind starts to blow with a velocity of $3\ \mathrm{m\ s^{-1}}$ due north. The relative velocity of the bird with respect to wind from east toward south is -

(A) $\tan^{-1}\left(\frac{3}{4}\right)$

(B) $\tan^{-1}\left(\frac{4}{3}\right)$

@IITJEE_Advanced

(C) $\tan^{-1}\left(\frac{5}{4}\right)$

(D) $\tan^{-1}\left(\frac{4}{5}\right)$

Q.7

A man can row a boat with $4\ \mathrm{km/h}$ in still water. If he is crossing a river where the current is $2\ \mathrm{km/h}$. Width of river is $4\ \mathrm{km}$. In what direction should he head the boat if he wants to cross the river in shortest time and what is this minimum time?

(A) Parallel to river current, $2\mathrm{hrs}$

(B) Perpendicular to river current, 2 hrs

(C) Parallel to river current, $1\mathrm{hrs}$

(D) Perpendicular to river current, 1 hrs

Q.8

A man can swim at the rate of $5\mathrm{kmh}^{-1}$ in still water. A $1-\mathrm{km}$ wide river flows at the rate of $3\ \mathrm{km\ h}^{-1}$. The man wishes to swim across the river directly opposite to the starting point,
How much time will he take to cross the river?

(A) 10 min

(B) 13 min

(C) 15 min

(D) 18 min

Q.9

A boat moves relative to water with a velocity v which is n times less than the river flow velocity u. At what angle to the stream direction must the boat move to minimize drifting?

(A) $\sin^{-1}\frac{1}{n}$

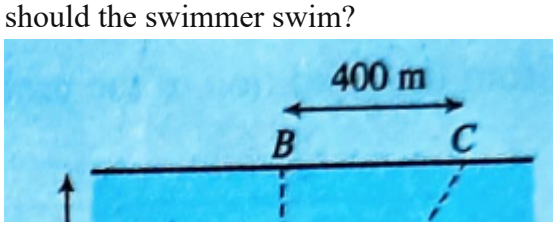
(B) $\cos^{-1}\frac{1}{1-n^{21/2}}$

(C) $\tan^{-1}\frac{1}{1-n^{21/2}}$

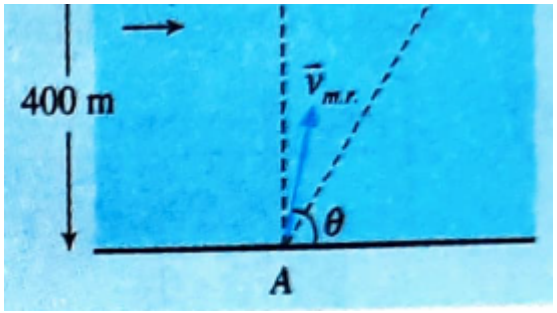
(D) $\sin^{-1}\frac{n^2-1^{1/2}}{n}$

Q.10

A river is flowing with a speed of $1\ \mathrm{kmh}^{-1}$. A swimmer wants to go to point C starting from A. He swims with a speed of 5 kmh^{-1} at an angle θ w.r.t. the river flow. If $AB = BC = 400\ \mathrm{m}$, at what angle with the river bank should the swimmer swim?



@IITJEE_Advanced



(A) 53°

(B) 30°

(C) 64°

(D) 74°

Q.11

A standing man observes rain falling with the velocity of 20 ms^{-1} at an angle of 30° with the vertical. Find the velocity with which the man should move so that rain appears to fall vertically to him.

(A) 15 m/s

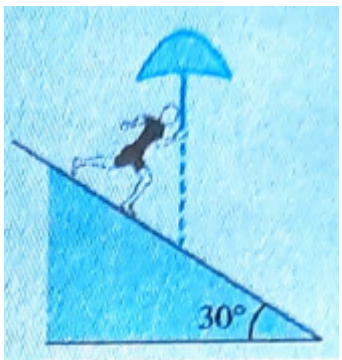
(B) 30 m/s

(C) 10 m/s

(D) 20 m/s

Q.12

A man is coming down an incline of angle 30°. When he walking with speed $2\sqrt{3}\ \mathrm{m\ s}^{-1}$ he has to keep his umbrella vertical to protect himself from rain. The actual speed of rain is $5\mathrm{m/s}$. At what angle with vertical should he keep his umbrella when he is at rest so that he does not get drenched?



(A) 58°

(B) 37°

(C) 42°

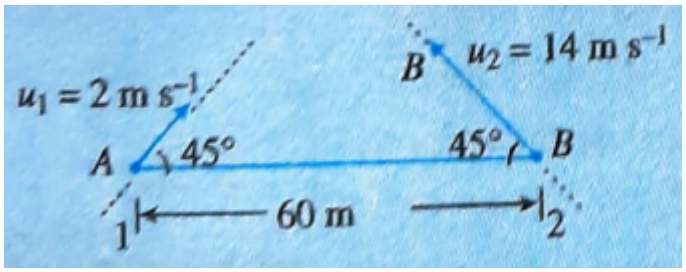
(D) 63°

Q.13

Two particles are located on a horizontal plane at a distance $60\ \mathrm{m}$. At $t = 0$ both the particles are simultaneously

@IITJEE_Advanced

projected at angle 45° with velocities $2\ \text{ms}^{-1}$ and $14\ \text{ms}^{-1}$, respectively.



Find the minimum separation between them during motion.

(A) 52 m

(B) 43 m

(C) 24 m

(D) 36 m

Q.14

Two cars A and B are moving west to east and south to north, respectively, along crossroads. A moves with a speed of 20 ms^{-1} and is $500\ \text{m}$ away from the point of intersection of cross roads and B moves with a speed of $15\ \text{m s}^{-1}$ and is $400\ \text{m}$ away from the point of intersection of cross roads. Find the shortest distance between them.

(A) 10m

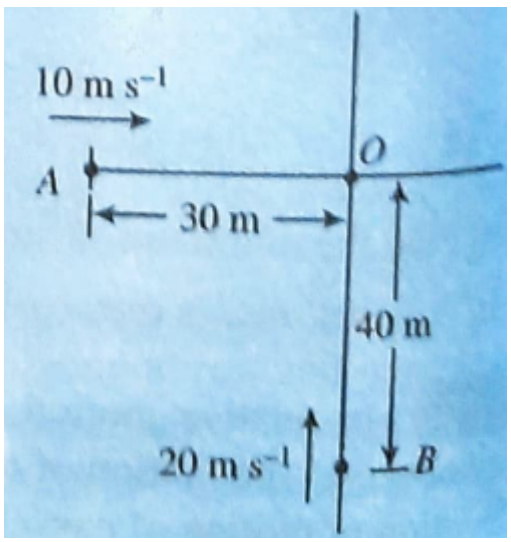
(B) 15m

(C) 20m

(D) 25m

Q.15

Two particles A and B are moving with uniform velocity as shown in figure given below at $t = 0$.
Find out the shortest distance between two particles.



(A) $= 4\sqrt{5}\ \text{m}$

@IITJEE_Advanced

(B) $= 5\sqrt{5}\ \mathrm{m}$

(C) $= 4\sqrt{4}\ \mathrm{m}$

(D) $= 5\sqrt{4}\ \mathrm{m}$

Q.16

An airplane is observed by two observers traveling at 60 $\mathrm{km/hour}$ in two vehicles moving in opposite directions on a straight road. To an observer, in one vehicle, the plane appears to cross the road track at right angles while to the other, it appears to be 45°. At what angle does the plane actually cross the road track and what is its speed relative to ground?

(A) $\tan^{-1} 2$

(B) $\tan^{-1} 1/2$

(C) $\tan^{-1} 3/2$

(D) $\tan^{-1} 2/3$

Q.17

A man wants to cross a river $500\ \mathrm{m}$ wide. The rowing speed of the man relative to water is $3\ \mathrm{km/hr}$ and river flows at the speed of $2\ \mathrm{km/hr}$. If man's walking speed on the shore is $5\ \mathrm{km/hr}$. then in which direction he should start rowing in order to reach the directly opposite point on the other bank in shortest time.

(A) $\sin^{-1}\left(\frac{7}{8}\right)$

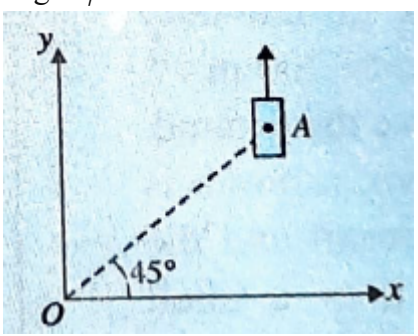
(B) $\sin^{-1}\left(\frac{8}{7}\right)$

(C) $\sin^{-1}\left(\frac{7}{3}\right)$

(D) $\sin^{-1}\left(\frac{3}{7}\right)$

Q.18

On a frictionless horizontal surface, assumed to be the $x-y$ plane, a small trolley A is moving along a straight line parallel to the y-axis with a constant velocity of $(\sqrt{3}-1)\mathrm{ms}^{-1}$. At a particular instant, when the line OA makes an angle of 45° with the x-axis, a ball is thrown along the surface from the origin O. Its velocity makes an angle ϕ with the x-axis and it hits the trolley.



The motion of the ball is observed from the frame of the trolley. Calculate the angle θ made by the velocity vector of the ball with the x-axis in this frame.

@IITJEE_Advanced

(A) 45°

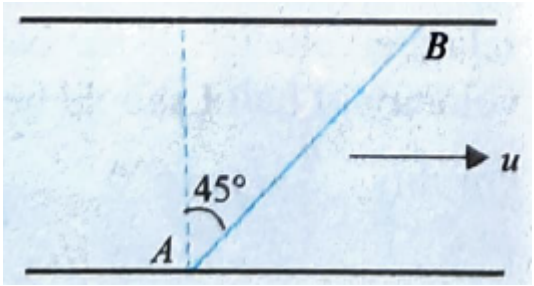
(B) 30°

(C) 60°

(D) 90°

Q.19

A man wants to reach point B on the opposite bank of a river flowing at a speed as shown in the figure. What minimum speed relative to water should the man have so that he can reach directly to point B ? In which direction should he swim?



(A) 45°

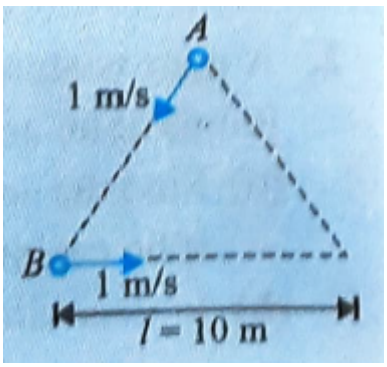
(B) 30°

(C) 60°

(D) 90°

Q.20

At $t = 0$, two particles A and B start moving with same speed along the sides of an equilateral triangle as shown in figure. Find the velocity of approach of A W.r.t. B at time $t = 5\text{sec}$



(A) 1

(B) 0

(C) 2

(D) 3

Answers & Solutions

Q.1 Answer:

$\frac{v}{a}\cos\alpha$

@IITJEE_Advanced

Solution:

$V_B = a \times t \qquad V_A = u$

$V_{BA} = at + (-u)$

$V_{BA}^2 = a^2t^2 + u^2 - 2au + \cos\alpha$

For maximum and minimum $\frac{d(v_{BA}^2)}{dt} = 0$

$2a^2t = 2au\cos\alpha$

$t = \frac{u\cos\alpha}{a}$

Q.2 Answer:

$\sqrt{a_1a_2}$

Solution:

Let s be the distance travelled by each car

$\sqrt{2a_1s} - \sqrt{2a_2s} = v_0$ **and** $\sqrt{\frac{2s}{a_2}} - \sqrt{\frac{2s}{a_1}} = t_0$

$\frac{v_0}{t_0} = \frac{\sqrt{a_1}-\sqrt{a_2}}{\frac{1}{\sqrt{a_2}}-\frac{1}{\sqrt{a_1}}} = \sqrt{a_1a_2}$

$\therefore$

Q.3 Answer:

$10\sqrt{3}\ ms^{-1}, 20\ ms^{-1}$

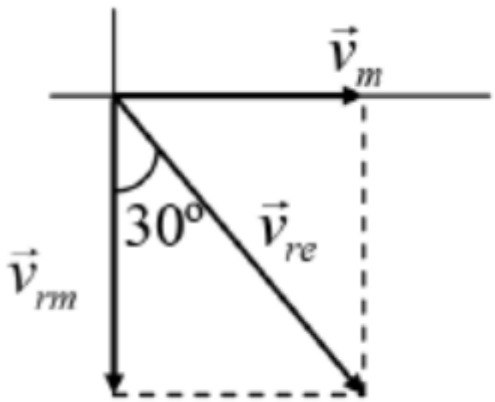
Solution:

Velocity of man $|\vec{v}_m| = 10ms^{-1}$

Using $\sin 30^\circ = \frac{v_m}{v_{re}}$

w.r.t.

$v_{ne} = \frac{v_m}{\sin 30} = \frac{10}{1/2}$



v_m = velocity of man
v_{re} = velocity of rain w.r.t. earth
v_{rm} = velocity of rain w.r.t. man

$= 20\ ms^{-1}$

Again $\cos 30^\circ = \frac{v_{ma}}{v_{re}}$

or $v_m = v_{re}\cos 30$

$= 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3}ms^{-1}$

$\therefore$

@IITJEE_Advanced

Q.4 Answer:

$v_B = u + v_A$

Solution:

At any instant of time, let the length of the string $BP = l_1$ and the length $PA = l_2$. In a further time t, let B move to the right by x and A move down by y, while P moves to the right by ut. As the length of the string must remain constant. $l_1 + l_2 = (l_1 - x + ut) + (l_2 + y)$

or $x = ut + y$

or $\dot{x} = u + y$

$\dot{x}$ = **speed of** B **to the right** $= v_B$, $\dot{y}$ = **downward speed of** $A = v_A$

$\therefore \quad v_n = u + v_A$

Also $\dot{v}_s = \dot{v}_\Delta$ **or** $a_B = a_A$

$\therefore$

Q.5 Answer:

120°

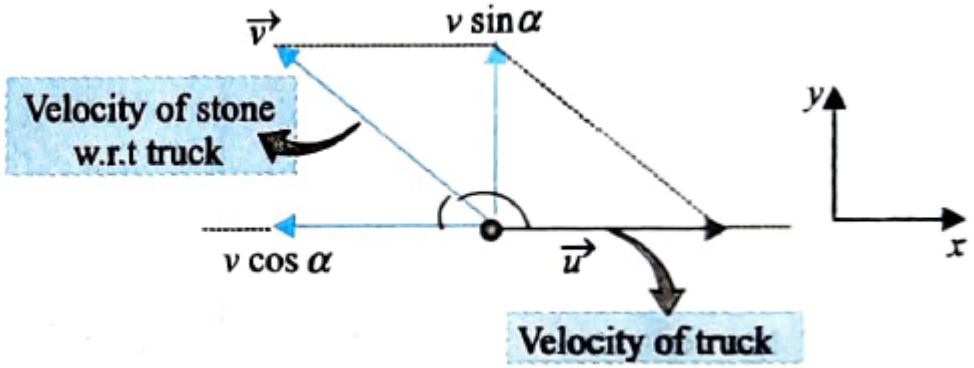
Solution:

The velocity of stone relative to truck, $\vec{v} = (-v\cos\alpha\hat{i} + v\sin\alpha\hat{j})$

The velocity of truck, $\vec{u} = u\hat{i}$ **The velocity of stone relative to the person G standing on ground**

$\vec{v}_{\text{stonc}} = \vec{v}_{\text{stone,truck}} + \vec{v}_{\text{truck}} = \vec{v} + \vec{u} = (-v\cos\alpha\hat{i} + v\sin\alpha\hat{j}) + u\hat{i}$

$\Rightarrow \quad \vec{v}_{\text{stone}} = (u - v\cos\alpha)\hat{i} + v\sin\alpha\hat{j}$



For ground observer, the stone appears at right angles to the truck. It means $u - v\cos\alpha = 0$

$u = v\cos\alpha$

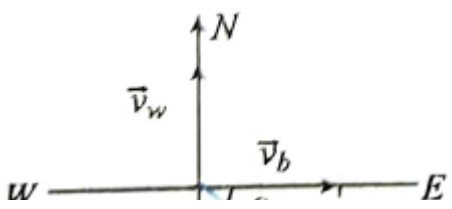
$\cos\alpha = \frac{u}{v} = \frac{15}{30} = \frac{1}{2} \Rightarrow \alpha = 60^\circ$ or $\theta = 120^\circ$

Q.6 Answer:

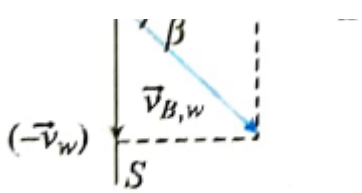
$\tan^{-1}\left(\frac{3}{4}\right)$

Solution:

The velocity of bird with respect to wind can be given as



@IITJEE_Advanced



$$\vec{v}_{b,w} = \vec{v}_b - \vec{v}_w = \vec{v}_b + (-\vec{v}_w)$$
$$= 4\hat{i} + (-3\hat{j})\,(\mathrm{ms}^{-1}) = 4\hat{i} - 3\hat{j}\ (\mathrm{m\ s}^{-1})$$
$$|\ \vec{v}_{b,w} = \sqrt{(4)^2 + (3)^2} = 5\ \mathrm{m\ s}^{-1}$$

Here the direction of the relative velocity of the bird is

$|\tan\beta| = \frac{3}{4} \Rightarrow \beta = \tan^{-1}\left(\frac{3}{4}\right)$

Hence, the relative velocity of the bird with respect to wind $5\ \mathrm{m\ s}^{-1}$ and in the direction $\tan^{-1}\left(\frac{3}{4}\right)$ from east toward south.

Q.7 Answer:

Perpendicular to river current, 1 hrs

Solution:

For the boat to cross the river in the shortest time, the man should head the boat in the direction perpendicular to the river current.

Given: Velocity of man wrt river, $\mathrm{V_{MW}} = 4\ \mathrm{km/hr}$ **Velocity of river,** $\mathrm{V_R} = 2\ \mathrm{km/hr}$

$\mathrm{d} = 4\ \mathrm{km}$

Minimum time taken to cross the river $\mathrm{t} = \frac{\mathrm{d}}{\mathrm{V_{MW}}} = \frac{4}{4} = 1\mathrm{hr}$

Q.8 Answer:

15 min

Solution:

Time taken to cross the river

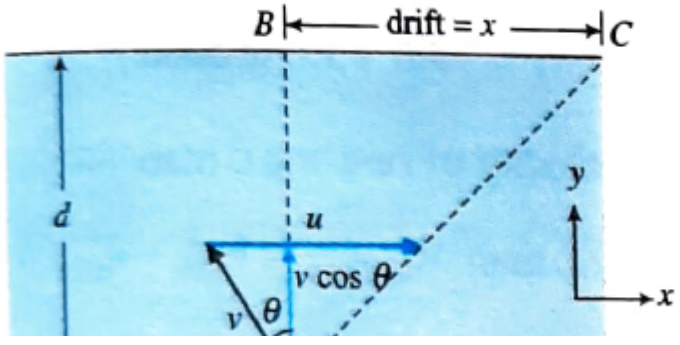
$t = \frac{d}{\sqrt{v^2 - u^2}} = \frac{1\ \mathrm{km}}{4\ \mathrm{km\ h}^{-1}} = \frac{1}{4}\ \mathrm{h} = 15\ \mathrm{min}$

Q.9 Answer:

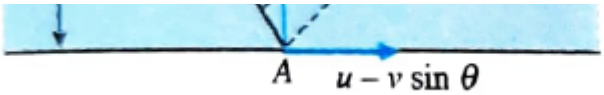
$\sin^{-1}\frac{1}{n}$

Solution:

In this case, the velocity of boat is less than the river flow velocity. Hence, boat cannot reach the point directly opposite to its starting point, i.e., drift can never be zero.



@IITJEE_Advanced



Suppose the boat starts at an angle θ from the normal direction up steam as shown in figure.

Component of the velocity of boat along the river,

$v_x = u - v\sin\theta$

and velocity perpendicular to the river, $v_y = v\cos\theta$.

Time taken to cross the river is $t = \frac{d}{v_y} = \frac{d}{v\cos\theta}$

Drift $x = (v_x)\,t = (u - v\sin\theta)\frac{d}{v\cos\theta} = \frac{ud}{v}\sec\theta - d\tan\theta$

The drift x **is minimum when** $\frac{dx}{d\theta} = 0$

or $\left(\frac{ud}{v}\right)(\sec\theta\cdot\tan\theta) - d\sec^2\theta = 0$

or $\frac{u}{v}\sin\theta = 1 \Rightarrow \sin\theta = \frac{v}{u}$

ie., for minimum drift, the boat must move at an angle $\theta = \sin^{-1}\theta = \sin^{-1}\left(\frac{v}{u}\right) = \sin^{-1}\frac{1}{n}$ **from the normal direction.**

Q.10 Answer:

53°

Solution:

Resultant path of the swimmer is at 45° **with bank, therefore,** x **- and** y**-components of swimmer's resultant velocity must be equal.**

Let velocity of swimmer $|\vec{v}_m| = v$

$\vec{v}_m = \vec{v}_{m,r} + \vec{v}_r$

$v_x = v_r + v_{m,r}\cos\theta$

$v_y = v_{m,r}\sin\theta$

Condition for reaching the point C**,**

$\tan 45^\circ = \frac{v_y}{v_x}, v_y = v_x$

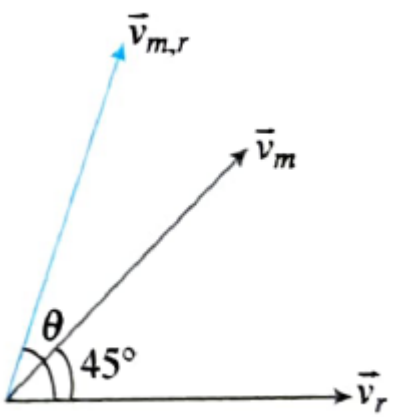
$(v_r + v_{m,r}\cos\theta) = v_{m,r}\sin\theta$

$1 + 5\cos\theta = 5\sin\theta$

On squaring, $1 + 25\cos^2\theta + 10\cos\theta = 25 - 25\cos^2\theta$

$50\cos^2\theta + 10\cos\theta - 24 = 0$

On solving, we get $\theta = 53^\circ$.



@IITJEE_Advanced

Q.11 Answer:

10 m/s

Solution:

Velocity of man, $\vec{v}_m = -v\hat{i}$ **(let)**

Velocity of rain, $\vec{v}_r = (-10\hat{i} - 10\sqrt{3}\hat{j})\text{ms}^{-1}$

Velocity of rain w.r.t to man, $\vec{v}_{r,m} = \vec{v}_r - \vec{v}_m$

$= (-10\hat{i} - 10\sqrt{3}\hat{j}) - (-v\hat{i})\text{ms}^{-1}$

$\vec{v}_{r,m} = -(10 - v)\hat{i} - 10\sqrt{3}\hat{j}\ \text{m s}^{-1}$

For vertical fall, the horizontal component must be zero.

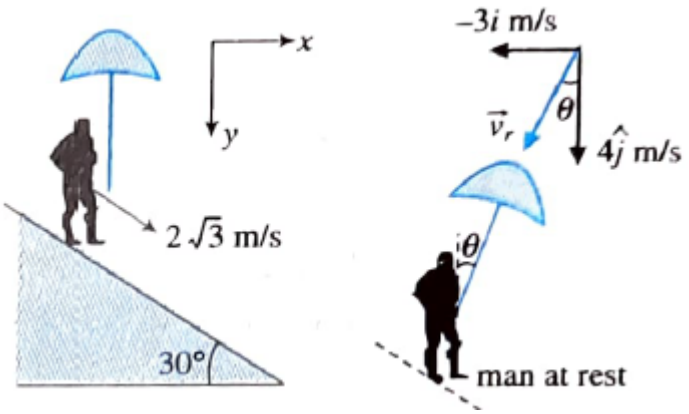
$\Rightarrow \quad -(10 - v) = 0$ **or** $v = 10\ \text{m s}^{-1}$

Q.12 Answer:

37°

Solution:

Velocity of rain w.r.t man



$\vec{v}_{r,m} = x\hat{j} = \vec{v}_r - \vec{v}_m$

$\vec{v}_m = 2\sqrt{3}[\cos 30\hat{i} + \sin 30\hat{j}] = 3\hat{i} + \sqrt{3}\hat{j}$

$\vec{v}_r = -3\hat{i} + (x - \sqrt{3})\hat{j}$

$5 = \sqrt{3^2 + (x - \sqrt{3})^2}$

$16 = (x - \sqrt{3})^2 \Rightarrow 4 + \sqrt{3} = x$

$\vec{v}_r = -3\hat{i} + 4\hat{j}$

$\tan\theta = \frac{3}{4} \Rightarrow \theta = 37^\circ$

Q.13 Answer:

36 m

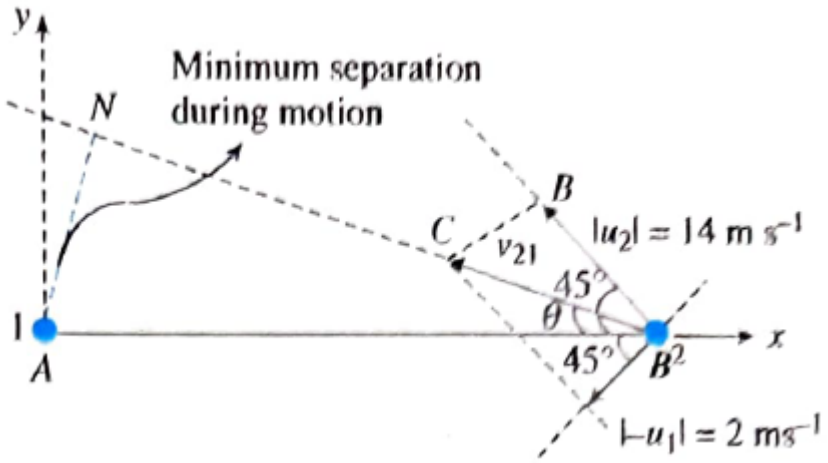
Solution:

In relative motion, observer considers himself at rest and observes the motion of object. Graphically, we can draw the direction of motion of particle 2 w.r.t. particle 1.

Both the particles are moving in gravitation field with same accelaeration g**. Hence, relative acceleration of particle 2 as seen from particle 1 will be zero. It means the relative velocity of the particle 2 w.r.t.**

@IITJEE_Advanced

particle 1 will be constant and will be equal to initial relative velocity. Graphically, we can draw the situation as shown in figure.



AN **is the minimum separation between the particles and** BN **is the relative separation between the particles when the distance between 1 and 2 is shortest. From figure, we can write**

$$v_{12}\cos\theta = 14\cos 45^\circ + 2\cos 45^\circ \quad \ldots(\text{i})$$

$$v_{12}\sin\theta = 14\sin 45^\circ - 2\sin 45^\circ \quad \ldots(\text{ii})$$

From (i) and (ii), $v_{12} = 10\sqrt{2}$ m s^{-1}

$\cos\theta = \frac{4}{5}$ and $\sin\theta = \frac{3}{5}$, as $\theta = 37^\circ$

Hence, minimum separation between the particles

$= AN = AB\sin\theta = 60 \times \frac{3}{5} = 36\text{m}$

Q.14 Answer:

20m

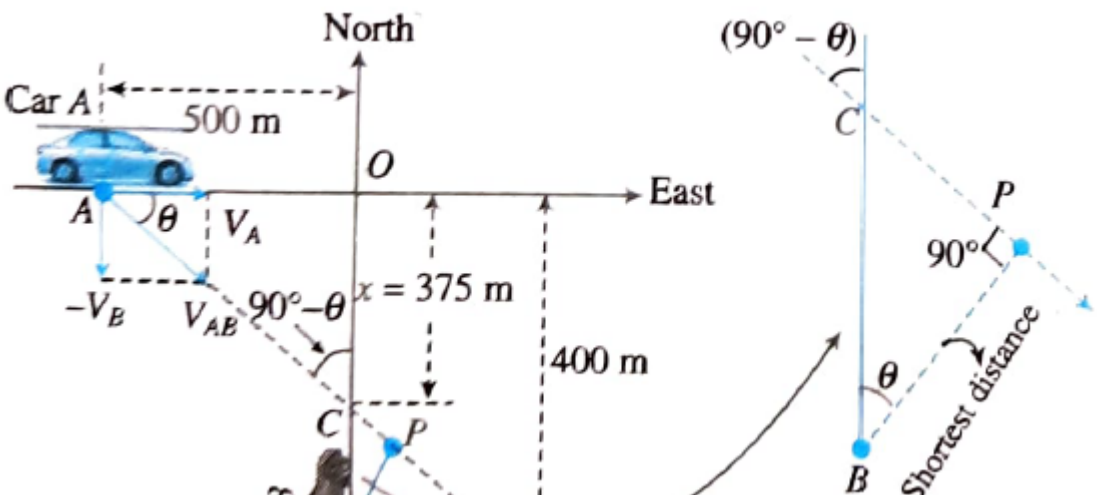
Solution:

Let us assume the velocity of A **w.r.t.** B**. To do this, we plot the resultant velocity,** $\vec{V}_{AB}$**.**

$$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B = \vec{V}_A + \left(-\vec{V}_B\right)$$

As the accelerations of both the cars is zero, so the relative acceleration between them is also zero. Hence, the relative velocity will remain constant. So the path of A **with respect to** B **will be straight line and along the direction of relative velocity of** A **with respect to** B**. The shortest distance between** A **and** B **is a perpendicular from** B **on the line of motion of** A **with respect to B. From figure,**

$\tan\theta = \frac{V_B}{V_A} = \frac{15}{20} = \frac{3}{4}$ **...(i)**



@IITJEE_Advanced



This θ is the angle made by the resultant velocity vector $\left|\vec{V}_{AB}\right|$ with the x-axis.

Let $OC = x$

Also we know that from figure, In $\triangle AOC$,

$\tan\theta = \frac{x}{500} = \frac{3}{4}$ **...(ii)**

From equation (i) and (ii), we get $x = 375$ m.

$\therefore \quad BC = OB - OC = 400 - 375 = 25$ m

But the shortest distance is BP.

From diagram, it is clear that

$BP = BC\cos\theta = 25 \times \frac{4}{5}$

$\therefore \quad BP = 20$ m

Q.15 Answer:

$= 4\sqrt{5}$ m

Solution:

Observing A **from the frame of** B

Let us draw $\vec{v}_{AB}$,

$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = \vec{v}_A + (-\vec{v}_B)$

$\tan\theta = \frac{10}{20} = \frac{1}{2}$

Again tan $\theta = \frac{AD}{CD} = \frac{AD}{40} = \frac{1}{2}$

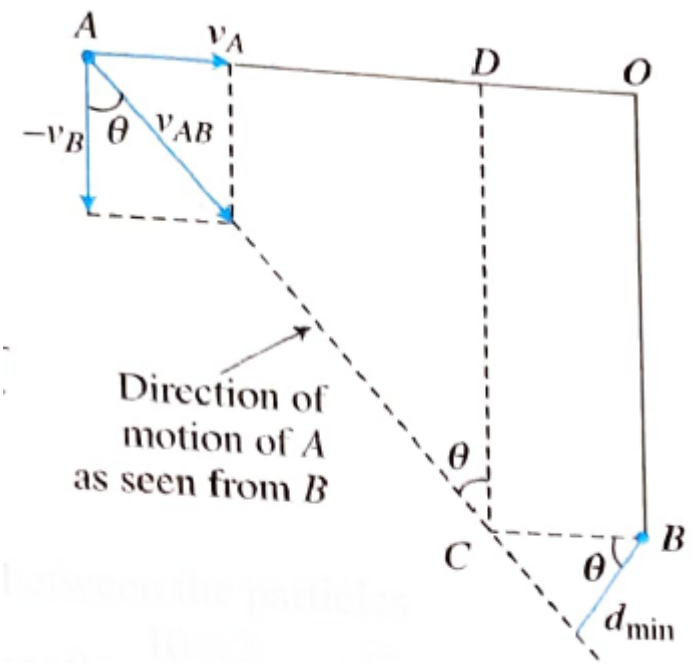
$\Rightarrow \quad AD = 20 \Rightarrow DO = 10$

$BC = 10$

Hence, the shortest distance between the particles

$d_{\text{short}} = BC\cos\theta = 10\cos\theta = \frac{10\times 2}{\sqrt{5}} = 4\sqrt{5}$ m

Since the closest distance is non-zero, therefore, the particles will not collide.



@IITJEE_Advanced

Q.16 Answer:

$\tan^{-1} 2$

Solution:

Let be $\vec{v}_p$ the velocity of plane relative to the ground, at angle θ to velocity $\vec{v}_1$ of observer in car 1.

In case (i), $\vec{v}_{p_1} = \vec{v}_p - \vec{v}_1$

$$\vec{v}_p = \vec{v}_{p_1} + \vec{v}_1$$

Vector diagram is shown in the figure. Note that according to observer in car 1 the plane crosses the road at right angles. Similarly, in case 2, $\vec{v}_p = \vec{v}_{p2} + \vec{v}_2$

We can combine figures (a) and (b) $\tan 45^\circ = \frac{AC}{AB}$

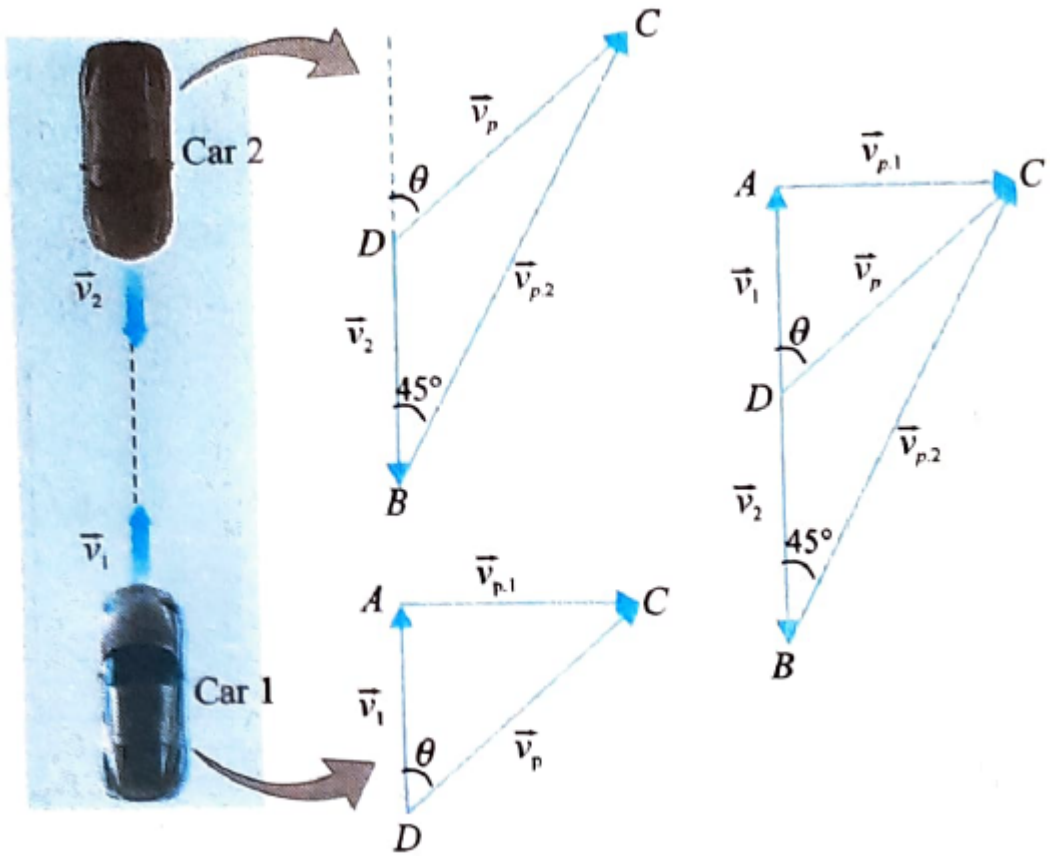
$$v_{P_1} = (v_1 + v_2)\tan 45^\circ = 120 \times 1 = 120 \text{ km/hr}$$

In $\triangle DAC, CD^2 = AD^2 + AC^2$

$v_p^2 = v_1^2 + v_{p,1}^2 = 60^2 + 120^2 \Rightarrow v_p = 60\sqrt{5}$ km/h

$\tan\theta = \frac{v_{p_1}}{v_1} = \frac{120}{60} = 2$

Hence $\theta = \tan^{-1} 2$



Q.17 Answer:

$\sin^{-1}\left(\frac{3}{7}\right)$

Solution:

Velocity of the man relative to ground,

$\vec{v}_m = (u - v\sin\theta)\hat{i} + v\cos\theta\hat{j}$

Hence, time taken by the man to cross the river is

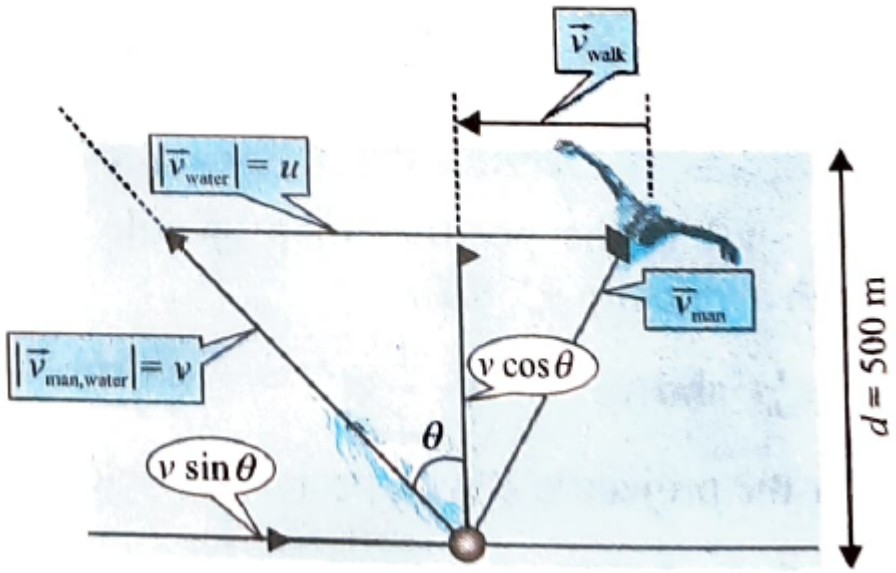
@IITJEE_Advanced

$t_1 = \frac{d}{v\cos\theta} = \frac{1}{2(v\cos\theta)}$

$\therefore$ **Drift of the man along the river is**

$$x = (u - v\sin\theta)t_1 = (u - v\sin\theta)\frac{1}{2(v\cos\theta)}$$

$$\Rightarrow \quad x = (2 - 3\sin\theta)\frac{1}{2(3\cos\theta)} = \frac{1}{3}\sec\theta - \frac{1}{2}\tan\theta$$



Time taken by the man to cover the distance,

$$t_2 = \frac{x}{v_{wull}} = \frac{\frac{1}{3}\sec\theta - \frac{1}{2}\tan\theta}{5} = \frac{1}{15}\sec\theta - \frac{1}{10}\tan\theta$$

$$t_1 = \frac{1}{2(v\cos\theta)} = \frac{\sec\theta}{6}, t_2 = \frac{1}{15}\sec\theta - \frac{1}{10}\tan\theta$$

Therefore, total time

$$T = t_1 + t_2 = \frac{1}{6}\sec\theta + \left(\frac{1}{15}\sec\theta - \frac{1}{10}\tan\theta\right)$$

$$\Rightarrow \quad T = \frac{7}{30}\sec\theta - \frac{1}{10}\tan\theta$$

for T to be minimum, $\frac{dT}{d\theta} = 0$

$$\frac{\pi T}{d\theta} = \frac{7}{30}\sec\theta\tan\theta - \frac{1}{10}\sec^2\theta = 0$$

$$\frac{7}{30}\sec\theta\tan\theta = \frac{1}{10}\sec^2\theta \Rightarrow \frac{7}{3}\tan\theta = \sec\theta$$

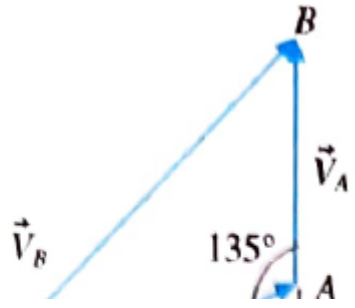
$$\sin\theta = \frac{3}{7} \Rightarrow \theta = \sin^{-1}\left(\frac{3}{7}\right)$$

Q.18 Answer:

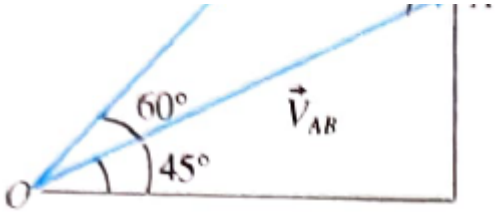
45°

Solution:

Since the ball hits the trolley, relative to trolley, the velocity of ball should be directed towards the trolley. Hence, in the frame of trolley, the ball will appear to be moving towards OA, or in the frame of trolley, ball's velocity will make an angle of 45°.



@IITJEE_Advanced

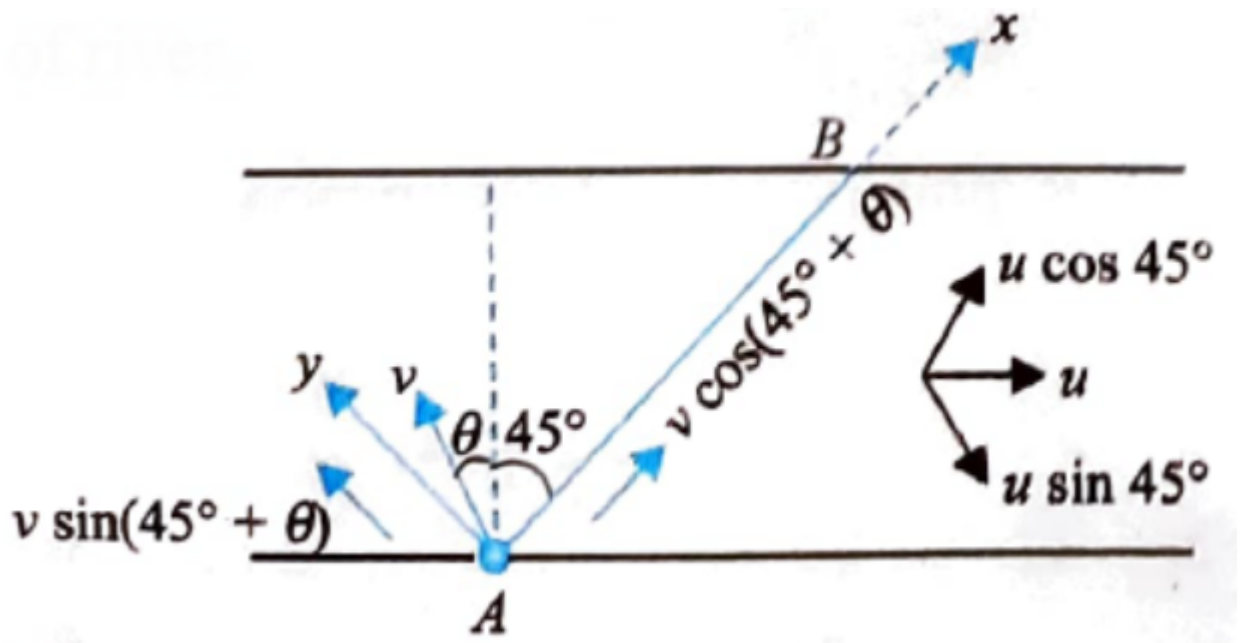


Q.19 Answer:

45°

Solution:

Let us consider the direction AB as x-axis and normal to AB line as y-axis. The man starts moving at angle θ from the line normal to the bank of river.



Velocity of the man w.r.t river

$\vec{v}_{m,w} = v\cos(45^\circ + \theta)\,\hat{i} + v\sin(45^\circ + \theta)\,\hat{j}$

Velocity of water $\vec{v}_w = u\cos 45^\circ \hat{i} - u\sin 45^\circ \hat{j}$

Hence velocity of the man, $\vec{v}_m = \vec{v}_{m,w} + \vec{v}_w\ \vec{v}_m = \left(v\cos(\theta + 45^\circ)\,\hat{i} + v\sin(\theta + 45^\circ)\,\hat{j}\right) + \left(u\cos 45^\circ \hat{i} - u\sin 45^\circ \hat{j}\right)$

$\vec{v}_m = (v\cos(\theta + 45^\circ) + u\cos 45^\circ)\,\hat{i} + (v\sin(\theta + 45^\circ) - u\sin 45\%)$

To reach point B, y-component of the man's velocity should be zero.

$v\sin(\theta + 45^\circ) = u\sin 45^\circ \Rightarrow v = \frac{v\sin 45^\circ}{\sin(\theta + 45^\circ)}$

For minimum speed of the man relative to water,

$\theta + 45^\circ = 90^\circ \Rightarrow \theta = 45^\circ$

Substituting $\theta = 45^\circ$ in (i), we get $v_{\min} = \frac{v\sin 45^\circ}{\sin(45^\circ + 45^\circ)} = \frac{u}{\sqrt{2}}$

Hence, the minimum speed relative to water should the man have so that he can reach directly to point B is $u/\sqrt{2}$, at angle 45° with upstream.

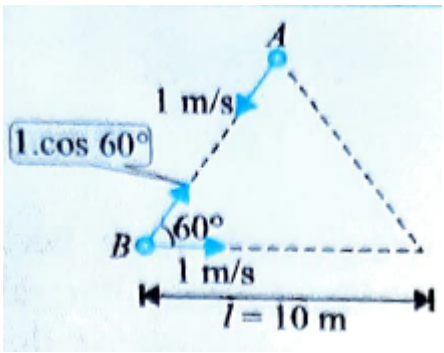
Q.20 Answer:

0

@IITJEE_Advanced

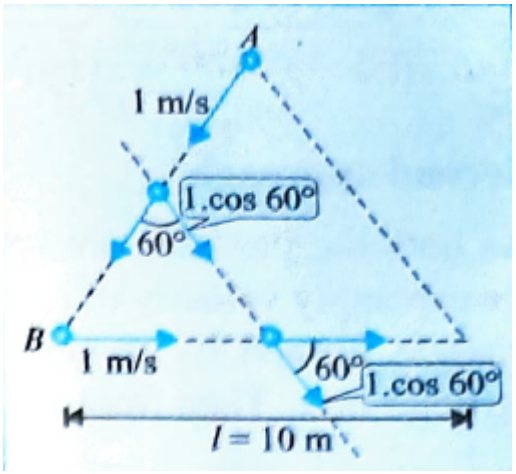
Solution:

(i) $t = 0$



Velocity of approach of A **w.r.t.** $B = 1 + 1 \cdot \cos 60^\circ = 1 + \frac{1}{2} = 1.5 \text{ m/s}$

(ii) $t = 5\text{sec}$



Velocity of approach of A **W.r.t.** B

$= 1.\cos 60^\circ - 1.\cos 60^\circ = 0$

@IITJEE_Advanced